RESEARCH PROJECT PAPER

E-COMMERCE SALES ANALYTICS DASHBOARD

An Interactive Data-Driven Framework for Sales Performance Monitoring and Business Decision Support

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Abstract

E-commerce businesses generate transactional records across orders, products, customers, payments, discounts, returns, and fulfilment. These records contain valuable evidence about sales performance, but raw tables are difficult to interpret consistently without a defined analytical process. This paper proposes an E-Commerce Sales Analytics Dashboard that transforms authorized transaction data into a unified set of business indicators and interactive visual views. The framework combines data ingestion, validation, cleaning, dimensional modelling, KPI computation, and dashboard presentation. It defines measures including gross merchandise sales, net sales, distinct orders, average order value, units sold, cancellation rate, and category contribution. Users can filter results by date, product category, region, order status, and payment method, then inspect trends and rankings through coordinated charts and tables. The paper contributes a reproducible project methodology, an explicit KPI dictionary, a data-quality strategy, and an evaluation plan covering calculation correctness, filter consistency, usability, and performance. No transaction dataset or measured prototype results were supplied during preparation; numerical examples are therefore illustrative, and no empirical business impact is claimed. The framework is intended to be implemented and evaluated using a public or institutionally authorized dataset.

Keywords: E-commerce analytics, business intelligence, sales dashboard, data visualization, KPI, data preprocessing, decision support.

1. Introduction

Digital commerce records every order and its associated product, customer, payment, and fulfilment information. These records can help a business understand which products contribute most to revenue, when demand changes, which regions generate sales, and how cancellations affect performance. However, insight does not emerge automatically from data collection. Transaction files may contain duplicate lines, missing values, inconsistent status labels, invalid dates, and ambiguous revenue definitions.

A dashboard can reduce the effort required to interpret such data, provided that it uses consistent business rules and clearly communicates each metric. A chart that counts order lines instead of distinct orders can overstate order volume. Similarly, a revenue total that ignores cancelled orders or refunds can misrepresent performance. This project therefore treats data quality and metric governance as core design requirements.

1.1 Problem Statement

Small and medium-sized online sellers may prepare sales summaries manually or use disconnected reports. This makes it difficult to compare periods, identify high-performing categories, and explain changes in sales indicators. The research problem is to design a practical, low-cost analytics workflow that converts transaction-level records into trustworthy and interactive sales information.

1.2 Aim and Objectives

Aim: To design an interactive dashboard that supports consistent monitoring and exploration of e-commerce sales data.

- Develop a repeatable process for importing, validating, cleaning, and transforming sales records.
- Create a documented KPI dictionary with explicit formulas and transaction eligibility rules.
- Design coordinated views for sales trends, products, categories, customers, geography, and payments.
- Enable filtering and comparison without changing underlying metric definitions.
- Evaluate the prototype using data-quality, correctness, usability, and performance tests.

1.3 Research Questions

1. How can raw e-commerce transaction data be prepared for reliable sales analysis?
2. Which KPIs and visual views provide a useful summary of sales performance?
3. How can validation rules and shared filters improve consistency across dashboard views?
4. What limitations arise when the framework is applied to incomplete or large datasets?

1.4 Scope

The core system focuses on descriptive and diagnostic analytics. It summarizes historical records and helps users explore patterns. Forecasting, recommendation, fraud detection, and causal inference are possible extensions rather than core claims of this project.

2. Background and Related Concepts

Business intelligence systems organize data so that users can monitor performance and investigate business questions. Dimensional modelling separates measurable business events from descriptive attributes, allowing consistent aggregation across time, product, customer, and geography [1]. Exploratory data analysis uses summaries and visual inspection to understand distributions, identify anomalies, and generate questions before formal modelling [2].

Visualization is most effective when the visual form matches the analytical task. Line charts are suitable for trends over time, bar charts support category comparisons, and tables expose exact values for inspection [3]. Python tools such as Pandas provide practical facilities for tabular cleaning and aggregation [4].

2.1 Project Contribution

This project does not claim a new statistical algorithm. Its applied contribution is a cohesive design that links transaction-grain documentation, transparent KPI definitions, logged data-quality treatment, and coordinated dashboard filters. These elements reduce the risk of visually attractive reports calculating similar metrics differently across pages.

3. Proposed System Architecture

The proposed architecture has four main stages: data sources, data preparation, an analytical model, and dashboard presentation. Each stage produces an inspectable output so that data and calculation issues can be traced.

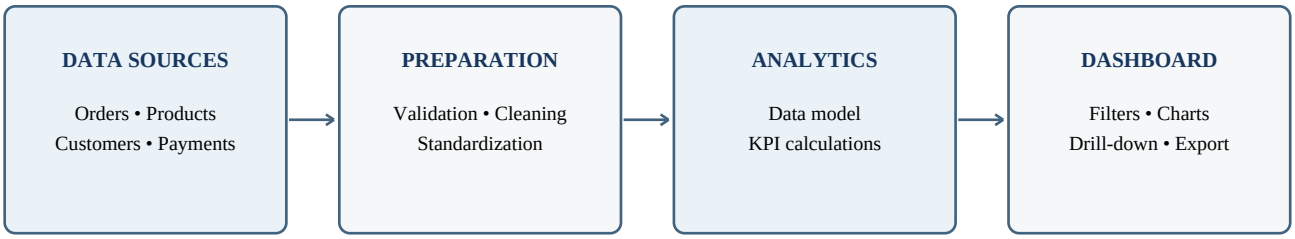

Figure 1. Proposed end-to-end architecture of the E-Commerce Sales Analytics Dashboard.

3.1 Data Sources

The system accepts CSV or Excel exports, or records stored in a relational database. Suitable entities include orders, order items, products, customers, payments, returns, and shipping records. The team should use a dataset that is public or explicitly authorized for academic use.

3.2 Data Preparation

The ingestion module checks required fields and data types. The cleaning module standardizes dates, currency, category names, and status values; detects duplicate records; validates numeric ranges; and creates a quality log. Original data should be preserved separately from transformed data to keep preprocessing reproducible.

3.3 Analytical Data Model

A recommended model is a star schema with one fact table and descriptive dimensions. The fact table stores measurable order-line events. Dimensions describe date, product, customer, geography, and payment method. The grain should be one row per order line unless the selected dataset requires another documented grain.

Table	Example fields	Purpose
fact_sales	order_id, line_id, order_date, product_id, customer_id, quantity, unit_price, discount, status	Stores order-line measures and keys.
dim_product	product_id, product_name, category, brand	Product and category analysis.
dim_date	date, day, month, quarter, year	Time-based grouping.
dim_customer	customer_key, segment (pseudonymous)	Privacy-conscious customer summaries.
dim_location/payment	region_key, region_name, payment_method	Region and payment comparisons.

Table 1. Suggested analytical data model.

4. Research Methodology

4.1 Research Design

The project follows a design-and-evaluation approach. First, business questions and data requirements are specified. Second, the data pipeline and dashboard are developed. Third, outputs are checked against manually verified calculations and representative user tasks. Finally, limitations and improvements are documented. This evaluates both analytical logic and usability.

4.2 Dataset Requirements

The dataset should contain an order identifier, order date, product identifier or category, quantity, unit price, and order status. Discount, refund, return, region, payment method, and pseudonymous customer identifier fields are desirable. The report should state the dataset source, license or permission, time coverage, currency, record count, and known missing fields.

4.3 Data Cleaning Procedure

1. Validate required columns, data types, and unique-key assumptions.
2. Measure missing values and flag records missing essential identifiers or dates.
3. Detect duplicate order-line keys and remove only confirmed duplicates, recording the action.
4. Standardize date formats, currency units, category labels, region names, and order statuses.
5. Check zero or negative quantities, invalid prices, impossible discounts, and conflicting return/cancellation states.
6. Review extreme values instead of deleting them automatically; bulk orders may be legitimate.
7. Retain before-and-after row counts and a log of excluded, corrected, or flagged records.

4.4 KPI Dictionary

Let i represent an eligible order line, q■ its quantity, p■ its unit price, d■ its discount, and r■ its refunded or returned value where available.

Metric	Formula / definition	Business meaning
Gross merchandise sales	GMS = Σ(q■ × p■)	Product value before adjustments.
Net sales	Net Sales = GMS – Σd■ – Σr■	Adjusted sales, if refund data is reliable.
Distinct orders	Count of distinct eligible order_id values	Orders, not order lines.
Average order value	AOV = Net Sales ÷ Distinct Orders	Adjusted sales per eligible order.
Units sold	Units = Σq■ for eligible lines	Total eligible quantity.
Cancellation rate	Cancelled orders ÷ placed orders × 100	Share of placed orders cancelled.
Category contribution	Category Net Sales ÷ Total Net Sales × 100	Category share of sales.

Table 2. KPI dictionary. Apply the same eligibility and status rules throughout the dashboard.

Metric policy: State how cancelled orders, partial returns, taxes, and shipping are treated. If refund or return values are unavailable, label the measure clearly; do not imply that it is fully refund-adjusted.

4.5 Analytical Techniques

- Descriptive statistics: totals, counts, averages, medians, and contribution percentages.
- Time-series summaries: daily, weekly, and monthly aggregation with period comparisons.
- Ranking: top products and categories by sales and units.
- Customer summaries: repeat-purchase frequency when stable pseudonymous customer keys exist.
- Segmentation: compare region, payment method, and category without claiming causation.

5. Dashboard Design and Functional Modules

5.1 Executive Overview

Show KPI cards for net sales, gross sales, distinct orders, average order value, units sold, and cancellation rate. Add a monthly trend and a comparison with the previous equivalent period.

5.2 Sales Trend Analysis

Use a line chart for daily, weekly, or monthly sales. Allow period comparison and clearly distinguish missing dates from zero-sales dates.

5.3 Product and Category Analysis

Use horizontal bar charts to rank products and categories by sales or units. Add a contribution chart and a detail table with product, category, units, orders, discounts, and net sales.

5.4 Customer, Region, and Payment Analysis

Where source data supports it, show new and returning customers, purchase frequency, region comparisons, and payment-method mix. These are descriptive summaries, not causal explanations.

5.5 Shared Filters and Export

Provide date, status, category, product, region, and payment filters. All charts must respond to the same filter state. Default exports to aggregated data and protect customer identifiers.

6. Implementation Strategy

A student prototype can use Python 3 and Pandas for preparation, SQLite or PostgreSQL for storage, and Streamlit or Power BI for presentation. Plotly or Matplotlib can provide charts. The final report should identify the actual tools and versions used.

Stage	Development task	Output
Requirements	Identify users, source fields, currency, date conventions, and business rules.	Requirements and KPI specification.
Profiling	Inspect counts, missing values, duplicates, status values, and ranges.	Dataset profile and quality report.
Preparation	Build repeatable cleaning and transformation scripts.	Cleaned dataset and audit log.
Analytics	Implement and test KPI functions or SQL views.	Reusable metric layer.
Dashboard	Build overview, trend, product, customer, and region pages.	Interactive dashboard.
Evaluation	Run validation, usability, and performance tests.	Test report and documentation.

Table 3. Recommended implementation stages.

6.1 Illustrative KPI Calculation

For an example order line with 2 units at ■500 each and a ■100 discount, gross sales are ■1,000 and sales after discount are ■900 before returns. If a period has ■90,000 adjusted sales across 100 eligible distinct orders, AOV is ■900. These values are hypothetical examples, not experimental findings.

7. Evaluation and Validation

Evaluate the prototype with a documented test dataset and a manually verified sample. Report the test environment, dataset size, test cases, outcomes, and unresolved defects.

Area	Test method	Evidence to report
Data quality	Check missing values, duplicate keys, invalid quantities, and statuses.	Counts and percentages before/after cleaning.
KPI correctness	Compare dashboard values with manual calculations.	Expected and actual values; mismatch count.
Filter consistency	Apply date, category, and region filters.	Pass/fail results for each view.
Usability	Ask users to find top category, trend, and AOV.	Task completion and feedback.
Performance	Measure data-load and refresh times.	Seconds, hardware, and record count.

Table 4. Evaluation matrix for the implemented prototype.

7.1 Example Test Cases

1. A duplicated order-line key should trigger the documented duplicate policy.
2. A cancelled order should be excluded from completed-sales KPIs if that is the declared rule.
3. An order with multiple product lines should count as one distinct order.
4. Changing a date filter should update all KPI cards and charts consistently.
5. If refund information is missing, the dashboard should display a limitation notice.

7.2 Reporting Actual Results

After implementation, insert the actual dataset title and source, date coverage, row count, distinct orders, missing-value rate, duplicate count, KPI validation outcomes, dashboard refresh time, and user-test observations. Results must be measured rather than estimated or invented.

8. Expected Results and Discussion

The expected outcome is a working prototype that consolidates sales indicators and enables users to explore performance through consistent filters. The overview is intended to reduce repeated manual preparation of basic sales summaries. Product and category pages should reveal contribution and ranking patterns. Time-series views make changes in sales visible, while customer and regional views provide additional context when the source data supports them.

Dashboard reliability depends on the entire analytical pipeline. A visually clear chart cannot correct a poorly defined metric or inaccurate source table. Therefore, the data-quality report, KPI dictionary, and validation tests should be delivered alongside the dashboard.

9. Privacy, Security, and Ethical Considerations

- Use public, synthetic, or institutionally authorized data; do not reuse private customer records without permission.
- Remove or pseudonymize direct identifiers before analytical processing.
- Restrict access to source files and limit dashboard exports to necessary information.
- Document dataset source, license, transformations, exclusions, and known limitations.
- Do not infer sensitive personal characteristics from dashboard summaries.
- Keep credentials outside source code and do not publish customer data in screenshots.

10. Limitations and Future Scope

The quality of the output is constrained by source-data completeness. Missing returns, refunds, taxes, or shipping values can affect revenue interpretation. Inconsistent product identifiers can fragment category totals. Customer analysis is not possible when stable customer keys are absent. Large datasets may require database-side aggregation, indexing, caching, or incremental refresh.

Future extensions may include demand forecasting, inventory alerts, customer segmentation, product recommendations, and anomaly detection. These should be added only after the descriptive dashboard is validated. Predictive models should use time-aware train/test splits, simple baselines, and suitable error metrics.

11. Conclusion

This paper proposed a practical framework for an E-Commerce Sales Analytics Dashboard integrating data preparation, dimensional modelling, transparent KPI definitions, and interactive visual analysis. The design supports sales monitoring across time, products, categories, customers, regions, and payment methods. Its central contribution is a reproducible workflow in which metric definitions, data-quality rules, and validation procedures are essential parts of the dashboard. The next step is to implement the system using an authorized dataset, conduct the evaluation plan, and replace illustrative examples with measured findings.

References

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Appendix A. Final Submission Checklist

- Insert the exact dataset title, source, license, and date accessed.
- Add screenshots of implemented dashboard pages and explain each chart.
- Include actual dataset profiling and cleaning results.
- Attach test cases with expected and actual KPI values.
- Report software versions, hardware, dataset size, and measured response times.
- Number and discuss every figure and table.
- Follow the university's required template, citation style, and originality rules.

Appendix B. Recommended Deliverables

Software: data import and cleaning script; analytical data model; KPI calculation module; interactive dashboard; requirements or environment file.

Documentation: dataset description; data dictionary; KPI dictionary; architecture diagram; test report; user guide; dashboard screenshots; final presentation.

Academic integrity note: This is an original project-specific research-paper draft prepared from the title and team details supplied. It is not a published or peer-reviewed article. The team must implement the prototype, cite the actual dataset, verify references, and add measured results before submitting it as completed empirical research.